Leonid Vinichenko

Senior Frontend Engineer, Expert 3D Graphics Software Engineer

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Summary: Developer with 10+ years of experience in frontend and 3D graphics (WebGL/WebGPU) in the browser. Responsible for architectural decisions, rendering system design, and team coordination. Focused on high performance and code quality.

Work Experience

Expert 3D Graphics Software Engineer at Align Technology

February 2024 – June 2025

Developer and architect of a 3D system for a web app used by orthodontists. Oversee architectural decisions and module integration, ensure code quality, and supervise part of the team.

- Drive architectural decisions and maintain coding standards
- Conduct regular code reviews, assign tasks, and mentor developers
- Coordinate **API** interaction with the backend, ensuring efficient two-way communication
- Leverage **WebGL**/**WebGPU** for high-performance **3D graphics**, writing custom shaders
- Designed and implemented an **Effect Compositor** system, integrating it into the **ECS** (**Entity-Component-System**). This significantly streamlined the creation of complex graphical effects and composite scenes.
- Implemented visual debugging tools that simplify the development process for junior engineers, accelerate feature delivery, and significantly reduce bugs.

Senior Frontend Engineer at Align Technology

July 2017 – January 2024

Developed a 3D editor for orthodontists (visualizing and editing jaw models). Contributed to UI, infrastructure, and WebSocket support.

- Built UI components using **React**, migrated project from ES6 to **TypeScript**
- Introduced **Redux** for state management, reducing technical debt
- Implemented runtime icon generation with **canvas 2D** for the 3D scene
- Created unit and integration tests (**Jest** + puppeteer)
- Managed backend communication via **WebSocket** (message queue, disconnection handling)
- Configured custom materials and shaders for **Three.js**, working with 3D math

Frontend Developer at Paragon Software

April 2015 – July 2017

Developed and maintained a single-page application (SPA) for the personal account system.

- Built the application with **AngularJS**, used responsive design techniques
- Exchanged data with the server via **RESTful API**
- Configured the build process using **Webpack**
- Developed a **Node.js** tool for localization (Excel → JSON)

Skills

JavaScript TypeScript ES6 React Redux AngularJS RxJS WebGL

WebGPU Three.js Shaders ECS Node.js RESTful API WebSocket

Webpack Jest Puppeteer Playwright HTML CSS/SASS 3D Math

Education

Peoples' Friendship University of Russia (RUDN)
2006 – 2010
Bachelor's degree in General Physics

Peoples' Friendship University of Russia (RUDN)
2010 – 2012
Master's degree in General Physics

Languages

English — Fluent